

3D Modeling, Animation & VFX with Blender

Day 1 – Orientation & Basics

- Meet peers, overview of program.
- Blender UI: viewport, outliner, properties, timeline.
- Navigation & object manipulation.
- Task: Arrange simple objects.

Day 2 – Modeling Fundamentals

- Edit mode: vertices, edges, faces.
- Extrude, loop cut, bevel.
- Modifiers: Mirror, Subdivision.
- Task: Model a scene with basic objects.

Day 3 – Intermediate Modeling

- Blocking → detail workflow.
- Proportional editing, snapping.
- Cleanup: normals, merging, shading.
- Good topology practices.
- Task: Create a themed interior scene.

Day 4 – Sculpting Basics

- Sculpt mode overview.
- Brushes: Grab, Clay, Smooth.
- Task: Sculpt an organic rock from reference.

Day 5 – UV Mapping & Texturing

- UV unwrap basics & good practices.
- Simple textures in Blender.
- Intro to Substance Painter.
- Task: Make a texture set for a model.

Day 6 – Materials & Shading

- Shader Editor intro.
- PBR: Base Color, Roughness, Normal.
- Glass, metal, wood, procedural textures.
- Task: Apply materials to Day 3 scene.

Day 7 – Lighting

- Light types: Point, Sun, Area, Spot.
- Three-point setup, HDRI.
- Color theory & mood.
- Task: Light a scene in 5 unique moods.

Day 8 – Rendering

- Eevee vs Cycles.
- Camera basics: Focal length, DOF.
- Render settings & optimization.
- Task: Render one complete scene.

Day 9 – Advanced Texturing

- Texture painting in Blender.
- Node-based shaders, masks, mixing.
- Task: Create an abstract material.

Day 10 – Animation Basics

- Keyframes: loc/rot/scale.
- Timeline, Dope Sheet, Graph Editor.
- Animate bouncing ball & camera flythrough.
- Task: Make a looping abstract animation.

Day 11 – VFX Intro

- Importing video footage.
- Compositing nodes.
- Motion tracking.
- Task: Add 3D object into real footage.

Day 12 – Particles & Simulations

- Particle systems: fire, smoke, sparks.
- Rigid body, fluid, cloth basics.
- Task: Create a sim-based render.

Day 13 – Final Project

- Apply skills from all modules.
- Deliverable: A **final render or short animation** showcasing modeling, texturing, shading, lighting, rendering, and (optionally) VFX or simulation techniques.